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DIPPER DOOR AND DIPPER DOOR TRIP ASSEMBLY

Abstract

A mining machine includes a boom, a handle coupled to the boom, a dipper coupled to the handle, and a dipper door pivotally coupled to the dipper. The mining shovel also includes a dipper door trip assembly including a trip motor coupled to the boom, a trip drum coupled to the handle, a linkage assembly coupled to the dipper door, a first actuation element extending directly from the trip motor to the trip drum, and a second actuation element extending directly from the trip drum to the linkage assembly.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. application Ser. No. 17/379,360, filed Jul. 19, 2021, which is a continuation of U.S. application Ser. No. 15/877,049, filed Jan. 22, 2018, which is a continuation of U.S. application Ser. No. 14/497,003, filed Sep. 25, 2014, and claims priority to U.S. Provisional Application No. 61/883,982, filed Sep. 27, 2013, and to U.S. Provisional Application No. 61/968,030, filed Mar. 20, 2014, the entire contents of each of which are incorporated herein by reference.

FIELD OF THE INVENTION

[0002] The present invention relates to the field of mining machines. Specifically, the present invention relates to a dipper door and a dipper door trip assembly on a mining machine, such as a rope shovel.

[0003] Industrial mining machines, such as electric rope or power shovels, draglines, etc., are used to execute digging operations to remove material from a bank of a mine. On a conventional rope shovel, a dipper is attached to a handle, and the dipper is supported by a cable, or rope, that passes over a boom sheave. The rope is secured to a bail that is pivotably coupled to the dipper. The handle is moved along a saddle block to maneuver a position of the dipper. During a hoist phase, the rope is reeled in by a winch in a base of the machine, lifting the dipper upward through the bank and liberating the material to be dug. To release the material disposed within the dipper, a dipper door is pivotally coupled to the dipper. When not latched to the dipper, the dipper door pivots away from a bottom of the dipper, thereby freeing the material out through a bottom of the dipper.

SUMMARY

[0004] In accordance with one construction, a mining shovel includes a boom, a handle coupled to the boom, a dipper coupled to the handle, and a dipper door pivotally coupled to the dipper. The mining shovel also includes a dipper door trip assembly including a trip motor coupled to the boom, a trip drum coupled to the handle, a linkage assembly coupled to the dipper door, a first actuation element extending directly from the trip motor to the trip drum, and a second actuation element extending directly from the trip drum to the linkage assembly.

[0005] In accordance with another construction, a dipper door trip assembly includes a trip motor, an actuation element coupled to the trip motor, and a linkage assembly coupled to the actuation element. The linkage assembly includes a lever arm coupled to the actuation element, a rod coupled to the lever arm about a first joint, a latch lever bar coupled to the rod about a second joint, and a latch bar coupled to the latch lever bar, wherein activation of the trip motor causes generally linear movement of the latch bar and latch bar insert, and wherein the first and second joints permit the rod to move in multiple degrees of freedom.

[0006] In accordance with another construction, a dipper door includes a bottom panel having a plurality of openings that open to an interior cavity inside the dipper door, a top panel, and a plurality of ribs extending between the bottom panel and the top panel.

[0007] Other aspects of the invention will become apparent by consideration of the detailed description and accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIG. **1** is a perspective view of a mining shovel.

[0009] FIG. **2** is a partial side view of a boom, handle, dipper, and dipper door of the mining shovel of FIG. **1**, as well as a dipper door trip assembly coupled to the shovel.

[0010] FIG. **3** is perspective view of a trip drum and actuation elements of the dipper door trip assembly.

[0011] FIG. **4** is a side view of an actuation element according to another construction.

[0012] FIG. **5** is a top view of the actuation element of FIG. **4**.

[0013] FIG. **6** is a side view of the actuation element of FIG. **4**, coupled to the trip drum.

[0014] FIGS. **7** and **8** are perspective views of the trip drum and actuation element of FIG. **4**, coupled to the linkage assembly.

[0015] FIG. **9** is a perspective view of the dipper door, and a linkage assembly of the dipper door trip assembly partially disposed within the dipper door.

[0016] FIG. **10** is a perspective view of the linkage assembly, with the dipper door removed.

[0017] FIG. **11** is an enlarged view of a lever arm of the linkage assembly partially disposed within the dipper door.

[0018] FIGS. **12** and **13** are enlarged views of a joint between the lever arm and a first end of a rod in the linkage assembly.

[0019] FIG. **14** is an enlarged view of a joint between a second end of the rod and a latch lever bar in the linkage assembly.

[0020] FIG. **15** is an enlarged view of a joint between the latch lever bar and a latch bar in the linkage assembly.

[0021] FIG. **16** is a view of the joint of FIG. **15**, with a housing element removed, illustrating an end of the latch lever bar.

[0022] FIG. **17** is a view of the joint of FIG. **15**, with the latch bar removed, illustrating an insert.

[0023] FIGS. **17**A and **17**B illustrate an embodiment of a cup roller assembly and latch a bar insert to be used with the linkage assembly.

[0024] FIG. **18** is a perspective view of the dipper door, illustrating openings and cavities sized to receive and hold the linkage assembly.

[0025] FIG. **19** is a section view of the dipper door, taken along lines **19-19** in FIG. **18**, illustrating a channel sized to receive and hold a portion of the linkage assembly.

[0026] FIG. **20** is a perspective view of the dipper door, illustrating a top panel of the dipper door.

[0027] FIG. **21** is a perspective view of the dipper door, illustrating a latch bar housing for the latch bar.

[0028] FIGS. **22** and **23** are perspective views of the dipper door, with a portion of the linkage assembly disposed therein.

[0029] FIG. **24** is a perspective view of an alternative design of the dipper door, illustrating openings and cavities sized to receive and hold the linkage assembly.

[0030] FIG. **25** is a section view of the dipper door, taken along lines **25-25** in FIG. **24**, illustrating a channel sized to receive and hold a portion of the linkage assembly.

[0031] FIG. **26** is a perspective view of the mining shovel, illustrating a channel on the dipper that receives a portion of a linkage assembly to latch a dipper door to the dipper.

[0032] Before any embodiments of the invention are explained in detail, it is to be understood that the invention is not limited in its application to the details of construction and the arrangement of components set forth in the following description or illustrated in the following drawings. The invention is capable of other embodiments and of being practiced or of being carried out in various ways. Also, it is to be understood that the phraseology and terminology used herein is for the purpose of description and should not be regarded as limited.

DETAILED DESCRIPTION

[0033] FIG. **1** illustrates a power shovel **10**. The shovel **10** includes a mobile base **15**, drive tracks **20**, a turntable **25**, a revolving frame **30**, a boom **35**, a lower end **40** of the boom **35** (also called a boom foot), an upper end **45** of the boom **35** (also called a boom point), tension cables **50**, a gantry tension member **55**, a gantry compression member **60**, a sheave **65** rotatably mounted on the upper end **45** of the boom **35**, a dipper **70**, a dipper door **75** pivotally coupled to the dipper **70**, a hoist rope **80**, a winch drum (not shown), a dipper handle **85**, a saddle block **90**, a shipper shaft **95**, and a transmission unit (also called a crowd drive, not shown). The rotational structure **25** allows rotation of the upper frame **30** relative to the lower base **15**. The turntable **25** defines a rotational axis **100** of the shovel **10**. The rotational axis **100** is perpendicular to a plane **105** defined by the base **15** and generally corresponds to a grade of the ground or support surface.

[0034] The mobile base **15** is supported by the drive tracks **20**. The mobile base **15** supports the turntable **25** and the revolving frame **30**. The turntable **25** is capable of 360-degrees of rotation relative to the mobile base **15**. The boom **35** is pivotally connected at the lower end **40** to the revolving frame **30**. The boom **35** is held in an upwardly and outwardly extending relation to the revolving frame **30** by the tension cables **50**, which are anchored to the gantry tension member **55** and the gantry compression member **60**. The gantry compression member **60** is mounted on the revolving frame **30**.

[0035] The dipper **70** is suspended from the boom **35** by the hoist rope **80**. The hoist rope **80** is wrapped over the sheave **65** and attached to the dipper **70** at a bail **110**. The hoist rope **80** is anchored to the winch drum (not shown) of the revolving frame **30**. The winch drum is driven by at least one electric motor (not shown) that incorporates a transmission unit (not shown). As the winch drum rotates, the hoist rope **80** is paid out to lower the dipper **70** or pulled in to raise the dipper **70**. The dipper handle **85** is also coupled to the dipper **70**. The dipper handle **85** is slidably supported in the saddle block **90**, and the saddle block **90** is pivotally mounted to the boom **35** at the shipper shaft **95**. The dipper handle **85** includes a rack and tooth formation thereon that engages a drive pinion (not shown) mounted in the saddle block **90**. The drive pinion is driven by an electric motor and transmission unit (not shown) to extend or retract the dipper handle **85** relative to the saddle block **90**.

[0036] An electrical power source (not shown) is mounted to the revolving frame **30** to provide power to a hoist electric motor (not shown) for driving the hoist drum, one or more crowd electric motors (not shown) for driving the crowd transmission unit, and one or more swing electric motors (not shown) for turning the turntable **25**. Each of the crowd, hoist, and swing motors is driven by its own motor controller, or is alternatively driven in response to control signals from a controller (not shown).

[0037] FIG. **2** illustrates a dipper door trip assembly **115** for the shovel **10**. The dipper door trip assembly **115** releases the dipper door **75** from the dipper **70** and allows the dipper door **75** to pivot away from a bottom of the dipper **70**. Although the dipper door trip assembly **115** is described in the context of the power shovel **10**, the dipper door trip assembly **115** can be applied to, performed by, or used in conjunction with a variety of industrial machines (e.g., draglines, shovels, tractors, etc.).

[0038] With reference to FIG. **2**, the dipper door trip assembly **115** includes a trip motor **120** disposed along the lower end **40** of the boom **35**. The trip motor **120** is powered by an electrical power source **122** (illustrated schematically) with its own motor controller. In some constructions the trip motor **120** is driven in response to control signals sent from a remotely located controller (e.g., a controller on the frame **30**).

[0039] With reference to FIGS. **2** and **3**, a first actuation element **125** (e.g., a wire rope, belt, or chain) is coupled to and extends directly from the trip motor **120** to a trip drum **130**. The trip drum **130** is releasably coupled to the dipper handle **85** with at least one mounting structure **135**, (e.g., a set of bolts and nuts), so that the trip drum **130** may be removed for repair or replaced with a different trip drum **130**.

[0040] As illustrated in FIG. **3**, the trip drum **130** includes a first drum portion **140** and a second drum portion **145**, both aligned along a common shaft **150** defining an axis of rotation **152**. The drum portion **140** is larger (e.g., in diameter) than the drum portion **145**, although in some constructions the drum portion **145** is larger than the drum portion **140**. The actuation element **125** is coupled to the drum portion **140** (e.g., fixed at one end of the actuation element **125** to the drum portion **140**), so that as the trip motor **120** turns, the actuation element **125** is either wound around or unwound from the drum portion **140**.

[0041] With reference to FIGS. **2** and **3**, a second actuation element **155** (e.g., a wire rope, belt, or chain) is coupled to and extends from the drum portion **145** directly to a linkage assembly **160**. The actuation element **155** is coupled to the drum portion **145** (e.g., fixed at one end of the actuation element **155** to the drum portion **145**), so that as the trip motor **120** turns, the actuation element **155** is either wound around or unwound from the drum portion **145**.

[0042] Because of the difference in size of the drum portions **140**, **145**, the trip drum **130** generates a mechanical advantage equivalent to the ratio of the diameter of the drum portion **140** to the diameter of the drum portion **145**. In some constructions, the ratio of the diameter of the drum portion **140** to the diameter of the drum portion **145** is greater than approximately 2.0. In some constructions, the ratio is between approximately 2.0 and 4.0. In some constructions, the ratio is greater than 3.0. Other constructions include different ranges and values.

[0043] The trip drum **130** advantageously removes the need for multiple sheaves, pulleys, or other structures to route the actuation elements **125**, **155** along the shovel **10**. Rather, as described above, the first actuation element **125** is routed directly from the trip motor **120** to the trip drum **130**, and the second actuation element **155** is routed directly from the trip drum **130** to the linkage assembly **160**.

[0044] The trip drum **130** also advantageously provides a reduction in whiplash effect generated during movement of the shovel **10**. Because the first and second actuation elements **125**, **155** are kept separate and are not directly coupled to one another, and because the trip drum **130** is heavy (e.g., at least 500 lbs.), any whiplash in the actuation element **125** (e.g., generated by rapid movement or swaying of the shovel **10**) will not substantially affect the movement and functionality of the actuation element **155**. Rather, a significant amount of inertia must be overcome in the trip drum **130** before the second actuation element **155** is affected negatively by any whiplash occurring in the actuation element **125**. In some constructions, the trip drum **130** also includes one or more dampers (e.g., linear or rotational) or friction disk brakes that further help to dampen any whiplash occurring in the actuation element **125**.

[0045] FIGS. **4-6** illustrate an actuation element **165** according to another construction. The actuation element **165** is a roller chain that allows for flat wind up and flat contact surface between the actuation element **165** and the drum **130**, similar to a sprocket, without chain twisting that can often cause wear. The life of the actuation element **165** is increased over traditional link chains (e.g., such as the actuation element **155** illustrated in FIG. **3**), particularly at the point where the actuation element **165** couples to the linkage assembly **160**, as well as where the actuation element **165** wraps around the drum **130**. The actuation element **165** provides improved wear characteristics for motion in the direction of rolling the chain onto the drum **130**. Reduction of wear and improvement of life at these points eliminates the need to constantly replace the actuation element **165**, which can occur every two weeks or sooner when a standard link chain is used as an actuation element. Less frequent replacement of the actuation element reduces the maintenance cost associated with the shovel **10**. In some constructions, the actuation element **165** lasts as long as nine to twelve months.

[0046] With reference to FIGS. **4** and **5**, in some constructions the actuation element **165** includes high strength end links **170**, as well as connectors **175** coupled to the end links **170**. The connectors

175 couple a first end **180** of the actuation element **165** to the drum **130** and a second end **185** of the actuation element **165** to the linkage assembly **160**. The connectors **175** include apertures **190** to couple the actuation element **165** to a pin or other structure on the drum **130** and the linkage assembly **160**. The end links **170** and the connectors **175** provide longer wear life where the actuation element **165** is coupled to the drum **130** and to the linkage assembly **160**. In some constructions, one or more of the end links **170** and the connectors **175** take all, or substantially all, of the wear in the actuation element **165** during use.

[0047] In some constructions the actuation element **165** is coupled to both a length of standard link chain and to the linkage assembly **160** in order to remove chain twist that causes wear at the drum **130**. In other constructions the actuation element **165** is coupled between two drums **130**, or between a drum **130** and another lever or linkage assembly in a mining machine other than the linkage assembly **160**.

[0048] With reference to FIGS. **7-12**, the linkage assembly **160** includes a lever arm **195** configured to be coupled to the actuation element **155** (or **165**). The lever arm **165** is disposed at least partially within the dipper door **75**, and is pivotally coupled to the dipper door **75** about a pivot structure **200**, such as a bolt or rod (FIGS. **11** and **12**) disposed in the dipper door **75**. As the actuation element **155** is moved by the trip motor **120**, the lever arm **195** is caused to pivot about the pivot structure **200**. Other constructions include different locations for the lever arm **195** than that illustrated, as well as different shapes and sizes. In some constructions the lever arm **195** is disposed substantially entirely within or entirely outside of the dipper door **75**.

[0049] With continued reference to FIGS. **9-12**, the linkage assembly **160** includes a further pivot structure **205**, such as a bolt or rod (FIGS. **11** and **12**) coupled to the lever arm **165**. The pivot structure **205** receives an end of the actuation element **155** (e.g., receives a link of a chain of the actuation element **155**, or the connector **175** in the case of the actuation element **165**), allowing the actuation element **155** to pivot relative to the lever arm **195** as the actuation element **155** is moved by the trip motor **120**. The pivot structure **205** is sized and shaped to absorb a substantial amount of stress generated by the pulling force of the actuation element **155** on the lever arm **195** as the actuation element **155** is moved by the trip motor **120**. The pivot structure **205** is easily removable from the lever arm **195** to be repaired or replaced.

[0050] With reference to FIGS. **10-14**, the linkage assembly **160** further includes a rod **210** pivotally coupled to the lever arm **195**. The rod **210** includes a first end **215** that is received at least partially within the lever arm **195** and pivots about a pivot structure **220** (including, e.g., a bolt or rod as illustrated in FIGS. **11** and **12**) coupled to the lever arm **195**, such that the rod **210** is able to pivot relative to the lever arm **195**. As illustrated in FIG. **13**, the rod **210** also includes a spherical bearing or bushing **225** within the first end **215**, thereby creating a spherical joint between the rod **210** and the lever arm **195** that permits freedom of movement and rotation of the rod **210** about multiple axes relative to the lever arm **195**. Other constructions include different types of joints between the rod **210** and the lever arm **195** (e.g., a ball joint, etc.).

[0051] With reference to FIGS. **10** and **14**, the rod **210** further includes a second end **230** that is coupled to a latch lever bar **235** of the linkage assembly **160**. As with the first end **215**, the second end **230** also includes a spherical bearing or bushing **240** that receives an end **244** of the latch lever bar **235**, thereby creating a spherical joint between the rod **210** and the latch lever bar **235** that permits freedom of movement and rotation of the rod **210** about multiple axes relative to the latch lever bar **235**. Other constructions include a different type of joint between the rod **210** and the latch lever bar **235** (e.g., a ball joint, etc.).

[0052] The use of spherical or ball joints between the rod **210** and both the lever arm **195** and the latch lever bar **235** permits deflections and adjustment of the rod **210** within the linkage assembly **160** during activation of the trip motor **120**. This freedom to move and deflect inhibits damage to the components of the linkage assembly **160**. While the illustrated construction utilizes spherical bearings or bushings **225**, **240** on the ends of the rod **210** to receive ends of the lever arm **195** and the latch lever bar **234**, in other constructions one or more of the spherical bearings or bushings are instead disposed on the lever arm **195** and/or the latch lever bar **235**, so as to receive ends of the rod **210**.

[0053] With reference to FIGS. **10** and **15-17**, the linkage assembly **160** further includes a latch bar **245** that is coupled to and receives the latch lever bar **235**. With reference to FIGS. **15-17**, the latch lever bar **235** passes through an opening **250** in the latch bar **245**. An insert **255** (e.g., metal) is disposed within an upper portion of the opening **250**. As illustrated in FIG. **17**, the insert **255** is coupled to the latch bar **245** with fasteners **260**. The insert **255** has a curved, contoured lower surface **265** that substantially matches a curved, contoured upper surface **270** on the latch lever bar **235**. The surfaces **265**, **270** act as bearing surfaces to permit some rotation and relative movement in at least one degree of freedom between the insert **255** and the latch lever bar **234**, thereby inhibiting wear and unwanted stress from damaging the linkage assembly **160**. The insert **255** prevents or inhibits wear of the latch bar **245**, and is easily removable and replaceable. In some constructions no insert **255** is provided. Rather, an inner surface of the latch bar **245** inside the opening **250** has a curved, contoured surface similar to the surface **265**.

[0054] With continued reference to FIGS. **15** and **16**, the linkage assembly **160** further includes a housing and pin assembly **272** that receives an end **275** of the latch lever bar **235** and permits movement of the end **275** in at least one degree of freedom (e.g., linearly). In the illustrated construction the housing and pin assembly **272** includes a carrier **280** that is shaped to receive the end **275**. The carrier **280** includes a curved, contoured surface **285** (FIG. **16**) that substantially matches a curved, contoured surface **290** on the latch lever bar **235**. The surface **285** retains the end **275** within the housing **280**. The housing and pin assembly **272** further includes a pin **295** that extends through an aperture **300** in an outer housing **305** and an aperture **302** in the carrier **280**. The carrier **280** is able to move (e.g., slide) along the pin **295** within the outer housing **305**, carrying the end **275** of the latch lever bar **235**. In some constructions the pin **295** and/or outer housing **305** is coupled to (e.g., affixed) the dipper door **75**, such that when the latch lever bar **235** is moved by rod **210**, the carrier **280** and the end **275** of the latch lever bar **235** are moved along a linear direction within the housing **305**, causing the latch bar **245** also to generally move along a linear direction.

[0055] In some constructions other structures are used to create one or more bearing surfaces for the latch lever bar **235**, and to facilitate movement of the latch lever bar **235** without damaging the latch bar **245**. For example, with reference to FIGS. **17**A and **17**B, in some constructions a cup roller assembly **306** is used, which includes a pin **307** and a roller **308** that rotates about the pin **307**. The pin **307** and the roller **308** are both coupled to and disposed at least partially within the latch bar **245**. The roller **308** engages a curved, contoured upper surface of the latch lever bar **235**. In the embodiment illustrated in FIGS. **17**A and **17**B, the latch lever bar **235** further includes a second roller **309** that is coupled to the carrier **280** and to the latch lever bar **235** to facilitate rotational movement of the end **275** of the latch lever bar **235**.

[0056] With reference to FIGS. **15** and **16**, the latch bar **245** includes an engagement portion **310** that facilitates easy gripping and/or removal of the latch bar **245** from the linkage assembly **160** to repair or replace the latch bar **245**. In the illustrated construction the engagement portion **310** is a recessed flange **315** with an aperture **320** that may receive a pin or other lifting structure that engages the engagement portion **310**. In other constructions the engagement portion **310** is a protruding flange with an aperture, or is another structure that allows easy gripping and removal of the latch bar **245** when desired.

[0057] With reference to FIGS. **9** and **10**, the linkage assembly **160** further includes a latch bar insert **325** disposed at an end of the latch bar **245**. In some constructions the latch bar insert **325** is formed as part of the latch bar **245**. The latch bar insert **325** extends from the housing dipper door **75**, and is moved along with the latch bar **215** when the trip motor **120** is activated and the actuation element **155** is moved. In the illustrated construction the latch bar insert **325** is a metal piece that absorbs stress imparted on the latch bar **245** during movement of the latch bar **245** into and out of engagement with the dipper **70**. The latch bar insert **325** is easily removed and replaced when damaged.

[0058] The linkage assembly **160** described above advantageously protects the life of its components. For example, and as described above, the second actuation element **155** (or **165**) is coupled directly to the pivot structure **205**, as opposed to the lever arm **195** itself. Therefore, if the pivot structure **205** fails, the pivot structure **205** can be replaced, without having to replace the entire lever arm **195**. Also, the spherical joints between the rod **210** and the lever arm **195** and the latch lever bar **235**, as well as the insert **255** (or other implemented bearing structure), increase the life of the linkage assembly **160** components by inhibiting wear and friction.

[0059] With reference to FIGS. **18-23**, the dipper door **75** includes panels, openings, channels, and cavities that receive and hold the linkage assembly **160** described above. In particular, the dipper door **75** includes a bottom panel **330** and a top panel **335**. The bottom panel **330** includes a front edge **340** and a rear edge **345**. The bottom panel **330** also includes openings **350** that open to an interior cavity **355** disposed inside the dipper door **75**. The openings **350** are spaced generally equally apart from one another along the bottom panel **330**. In the illustrated construction at least some of the openings **350** are generally disposed closer to the edge **340** than the edge **345**. Five openings **350** are illustrated, although in other constructions different numbers, sizes, shapes, and arrangements of openings **350** are used.

[0060] As illustrated in FIGS. **18**, **19**, **22**, and **23**, the openings **350** are elongate, and have first ends **360** and second ends **365**. The first ends **360** are disposed closer to the edge **345** than the second ends **365**. The second ends **365** of the openings **350** are arranged in a generally arched or curved pattern along the bottom panel **330**, such that the second ends **365** are aligned along a curved axis **370** that extends along the bottom panel **330**. Because at least some of the openings **350** are disposed closer to the edge **340** than the edge **345**, the bottom panel **330** includes a solid portion **375** between the curved axis **370** and the edge **345**. The solid portion **375** provides structural strength to the bottom panel **330** and to the dipper door **75**.

[0061] With continued reference to FIGS. **18**, **19**, **22**, and **23**, the dipper door **75** also includes ribs **380** that are disposed between the panels **330**, **335**. Some of the ribs **380** extend directly from the bottom panel **330** to the top panel **335**. The ribs **380** provide additional structural support for the dipper door **75** to accommodate for absent material in the openings **350** and the cavity **355**, and also help to evenly distribute loads within the dipper door **75** during an impact loading (e.g., when the dipper door **75** slams shut quickly against the dipper **70**). Use of the structural ribs **380** allows the top panel **335** to remain generally thin, helping to reduce the overall weight of the dipper door **75**, while still maintaining high strength for the dipper door **75**. As illustrated in FIGS. **18**, **19**, **22**, and **23**, some of the ribs **380** include openings **385** that are sized and shaped to receive, hold, and guide the latch lever bar **235** within the dipper door **75**.

[0062] With reference to FIG. **21**, the dipper door **75** further includes a latch bar housing **390** forming a channel **395** that extends from the interior cavity **355** to an exterior surface **400** of the dipper door **75**. In some constructions the latch bar housing **390** is integrally formed as one piece within the dipper door **75**. In some constructions the latch bar housing **390** is a separate piece. The channel **395** is sized and configured to receive the latch bar **245** and the latch bar insert **325**. In some constructions the latch bar housing **390** also includes one or more bearings or guide surfaces (e.g., plastic or nylon bearing inserts, roller bearings, other types of rollers, etc.) that facilitate sliding movement of the latch bar **245** within the latch bar housing **390**, and inhibit damage to the latch bar **245**.

[0063] With reference to FIGS. **18** and **22**, the dipper door **75** further includes an opening **405** in an arm **410** that receives at least a portion of the lever arm **195**, such that the lever arm **195** is disposed at least partially within the arm **410** of the dipper door **75**.

[0064] With reference to FIGS. **19** and **22**, the arm **410** forms a rectangular, box-like frame defining an interior channel **415** that extends toward the cavity **355**. The rod **210**, which is coupled to the lever arm **195**, extends through the channel **415** and into the cavity **355**, where the rod **210** is coupled to the latch lever bar **235**. The box-like structure of the arm **410** provides added structural support for the dipper door **75**.

[0065] With continued reference to FIGS. **18**, **19**, **22**, and **23**, dipper door **75** also includes webs **417**, **418** that are disposed between the openings **350**, the webs **418** being primary webs that are angled directly toward the arms **410**. The primary webs **418** absorb a significant amount of load and provide further added strength to the dipper door **75**. In some constructions the primary webs **418** provide a load path along the dipper door **75** that absorbs approximately at least 90% of a load acting on the dipper door **75**. In some constructions the primary webs **418** absorb between approximately at least 95% of a load acting on the dipper door **75**. Other constructions include different ranges.

[0066] The openings **350**, along with the cavity **355**, reduce the amount of material needed for the dipper door **75**, which makes the dipper door **75** more light-weight than conventional dipper doors. While the dipper door **75** is more light-weight than conventional dipper doors, in some constructions the dipper door **75** has equal (or even greater) overall structural strength than conventional dipper doors, due at least in part to the arrangement of the solid portion **375**, the ribs **380**, the box-like structure of the arms **410**, the webs **417** and **418**, and the top and bottom panels **345**, **340** overall.

[0067] FIGS. **24** and **25** illustrate an alternative construction of a dipper door **420**.

[0068] As illustrated in FIGS. **24** and **25**, elongate openings **450** are provided, similar to the openings **350**, the elongate openings **450** having first ends **460** and second ends **465**. Some of the first ends **460** are disposed closer to an edge **440** than the second ends **465**. In the illustrated construction of FIGS. **24** and **25**, both the first and second ends **460**, **465** are arranged in a generally arched or curved pattern along a bottom panel **430**, such that the second ends **465** are aligned along a curved axis **470**, and the first ends **460** are aligned along a curved axis **472**. In some constructions the curved axes **470**, **472** are parallel. The panel **430** includes a solid portion **475** between the curved axis **472** and the edge **445**.

[0069] With reference to FIGS. **24** and **25**, the dipper door **420** also includes ribs **480**, similar to ribs **380**, that are disposed between the panels **430**, **435** and include openings **485**, as well as a latch bar housing **490** and an opening **505** in an arm **510** that receives the lever arm **195**.

[0070] As illustrated in FIG. **25**, the dipper door **420** includes an interior channel **515** in the arm **510**, similar to the channel **415**. The dipper door **420** also includes two ribs **520** that extend through the channel **515** and guide the rod **210**. The two ribs **520** add further structural support within the arm **510**. As illustrated in FIGS. **25** and **26**, the rod **210** extends through the channel **515** and through an opening **525** into a cavity **455**, where the rod **210** is coupled to the latch lever bar **235**.

[0071] With reference to FIG. **26**, the dipper **70** includes a channel **460** disposed along a lower edge portion **465** of the dipper **70**. The channel **460** and the latch bar housing **490** (or **390** in the case of the dipper door **75**) are aligned with one another during a latched condition, such that the latch bar insert **325** extends through the latch bar housing **490**, **390** and at least partially into the channel **460**, thereby locking movement of the dipper door **420**, **75** relative to the dipper **70**.

[0072] With reference to FIGS. **1-26**, in order to release the dipper door **420**, **75** from the latched condition, the trip motor **120** is activated by the controller **122**. When the trip motor **120** is activated the trip motor **120** pulls the first actuation element **125** toward the trip motor **120**, thereby causing rotation of the drum portion **140** about the axis **152**. As the drum portion **140** rotates, the drum portion **145** also rotates about the axis **152**, causing the second actuation element **155** to be pulled toward and wound about the second drum portion **145**.

[0073] Movement of the second actuation element **155** causes the lever arm **195** to pivot relative to the pivot structure **200**, which causes the rod **210** to move (e.g., be pulled up through the opening **300**). As the rod **210** is moved, the spherical joints at the first end **215** and the second end **230** of the rod **210** permit relative rotational movement between the rod **210** and both the lever arm **195**

and the latch lever bar **235**, accounting for any pivoting and arching movement of the lever arm **195** about the pivot structure **200**.
[0074] As the rod **210** moves, the movement of the rod **210** generates a generally linear movement of the latch lever bar **235**, and the movement of the latch lever bar **235** generates a generally linear movement of the latch bar **245** within the latch bar housing **490**, **390** (e.g., with the guidance of the housing and pin assembly **272**). As the latch bar **245** is moved within the latch bar housing **490**, **390**, the latch bar insert **325** is pulled away from the dipper **70**, thereby freeing the dipper door **420**, **75** from the dipper **70**, and allowing the dipper door **420**, **75** to swing and pivot open relative to the bottom of the dipper **70** to unload material. As the material is unloaded, for example, into a truck or other vehicle, the components of the dipper door trip assembly **115** are positioned to remain well away from the truck and to not interfere with the unloading process.
[0075] To return the latch bar insert **325** back into the channel **460** after the material has been unloaded, gravity is used (i.e., the latch bar **245** is naturally urged toward the latched position by gravity). In other constructions, a biasing member or members are used to urge the latch bar **245** and the latch bar insert **325** toward the latched position. Because of the high mechanical advantages and forces possible with the dipper door trip assembly **115** described above, the latch bar insert **325** may be safely extended deep into the channel **460** during this latched condition. This results in a significantly lower likelihood of a false trip and release of the dipper door **420**, **75**.
[0076] With reference to FIG. **17**B, in some constructions the latch bar insert **325** includes a marking **495** (e.g., line, slot, groove, etc.) that aids in aligning the latch bar insert **325** within the latch bar housing **490**, **390** during installation or manufacturing of the dipper door **420**, **75**. For example, in some constructions the latch bar insert **325** is aligned (in a non-latching state) such that the marking **495** coincides with an outer surface (e.g., such as surface **400**) of the dipper door **400** or **75**, thereby providing an indication that the dipper door trip assembly **115** has been installed correctly. As illustrated in FIG. **17**B, in some constructions the latch bar insert **325** is installed with plurality of fasteners **496**.
[0077] In the event that the dipper door **420**, **75** slams quickly against the dipper **70** with high impact (e.g., because of a snubber failure) during the unloading process or during the process of the latch bar **325** returning to the latched position, the dipper door trip assembly **115** is able to absorb and withstand the impact without failing or incurring undesired wear. This is due at least in part to the spherical joints and contoured surfaces within the linkage assembly **160** described above. Similarly, the ribs **480**, **380** and webs **417**, **418** in the dipper door **420**, **75** are also able to absorb and withstand the impact without causing damage to the dipper door **420**, **75** or the linkage assembly **160** disposed within the dipper door **75**.
[0078] Although the invention has been described in detail with reference to certain preferred embodiments, variations and modifications exist within the scope and spirit of one or more independent aspects of the invention as described.

Claims

1. A dipper door for a mining machine, the dipper door comprising: a bottom panel; a top panel spaced from the bottom panel, wherein the bottom panel and the top panel define an interior cavity therebetween; a first rib positioned within the interior cavity; a second rib positioned within the interior cavity; a first dipper door arm extending away from the interior cavity and configured to pivotally couple to a dipper; and a second dipper door arm extending away from the interior cavity and configured to pivotally couple to the dipper.
2. The dipper door of claim 1, wherein the first rib extends from the first arm and the second rib extends from the second arm.
3. The dipper door of claim 1, wherein the first rib and the second rib are angled relative to one another at a non-zero angle.

4. The dipper door of claim 1, wherein the first rib includes a first opening sized and shaped to receive and guide a latch lever bar.
5. The dipper door of claim 1, wherein the first rib extends directly from the bottom panel to the top panel.
6. The dipper door of claim 1, further comprising a latch bar housing forming a channel that extends from the interior cavity to an exterior surface of the dipper door.
7. The dipper door of claim 1, wherein the first arm includes an opening configured to receive a portion of a latch lever arm.
8. The dipper door of claim 1, wherein the first arm includes a rectangular frame defining an interior channel that extends toward the interior cavity.
9. The dipper door of claim 1, further comprising a third rib positioned within a channel in the first arm, and a fourth rib positioned within the channel in the first arm, wherein the third rib and the fourth rib are configured to guide a rod of a dipper door trip assembly.
10. The dipper door of claim 1, wherein the bottom panel includes a front edge and a rear edge, wherein the bottom panel includes openings positioned between the front edge and the rear edge that open to the interior cavity.
11. The dipper door of claim 10, wherein the openings are positioned closer to the rear edge than the front edge.
12. The dipper door of claim 10, wherein the bottom panel includes webs extending between the openings.
13. The dipper door of claim 10, wherein each of the openings is elongate, and includes a first end and a second end.
14. The dipper door of claim 13, wherein the second ends of the openings are aligned along a curved axis that extends along the bottom panel.
15. The dipper door of claim 14, wherein the curved axis is a first curved axis, wherein the first ends of the openings are aligned along a second curved axis that extends along the bottom panel.
16. A dipper door for a mining machine, the dipper door comprising: a bottom panel; a top panel spaced from the bottom panel, wherein the bottom panel and the top panel define an interior cavity therebetween; wherein the bottom panel includes openings that open to the interior cavity, and wherein the bottom panel includes webs extending between the openings; a first dipper door arm extending away from the interior cavity and configured to pivotally couple to a dipper; and a second dipper door arm extending away from the interior cavity and configured to pivotally couple to the dipper.
17. The dipper door of claim 16, wherein the webs include a first web extending form the first arm, and a second web extending from the second arm, wherein the first web and the second web converge toward one another moving in a direction away from the first and second arms.
18. The dipper door of claim 16, wherein the bottom panel includes an end region, wherein the webs converge toward the end region, and wherein the end region extends to and couples to an end of the top panel.
19. The dipper door of claim 16, further comprising ribs positioned within the interior cavity, wherein at least one of the ribs includes an opening.
20. The dipper door of claim 19, wherein the ribs include a first rib and a second rib, wherein the webs include a first web and a second web, wherein the first rib extends from the first web to the top panel, and wherein the second rib extends from the second web to the top panel.